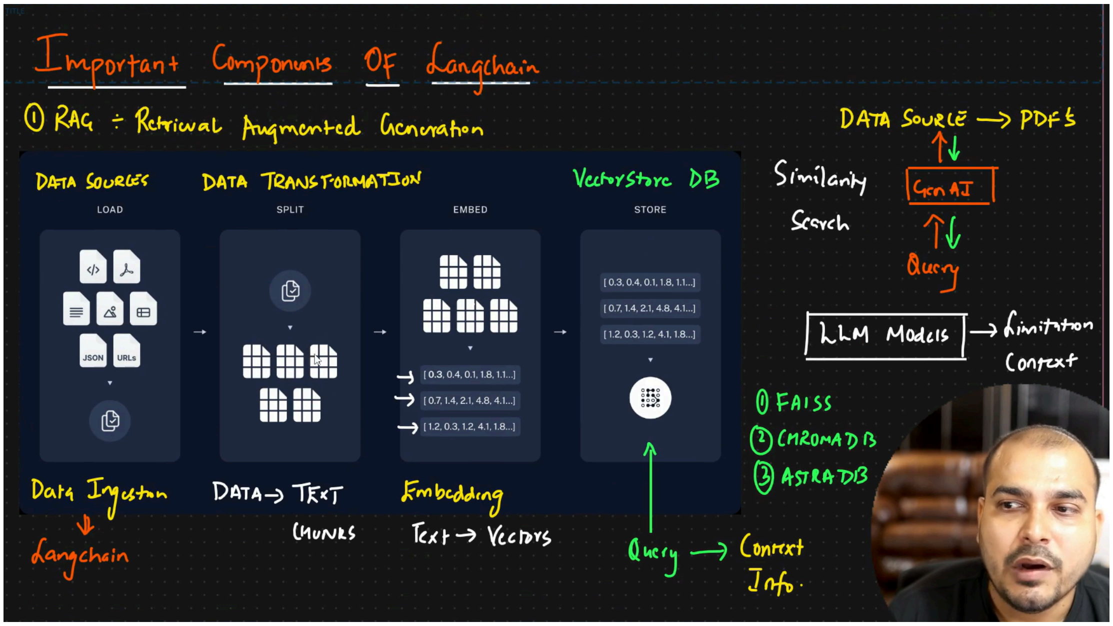
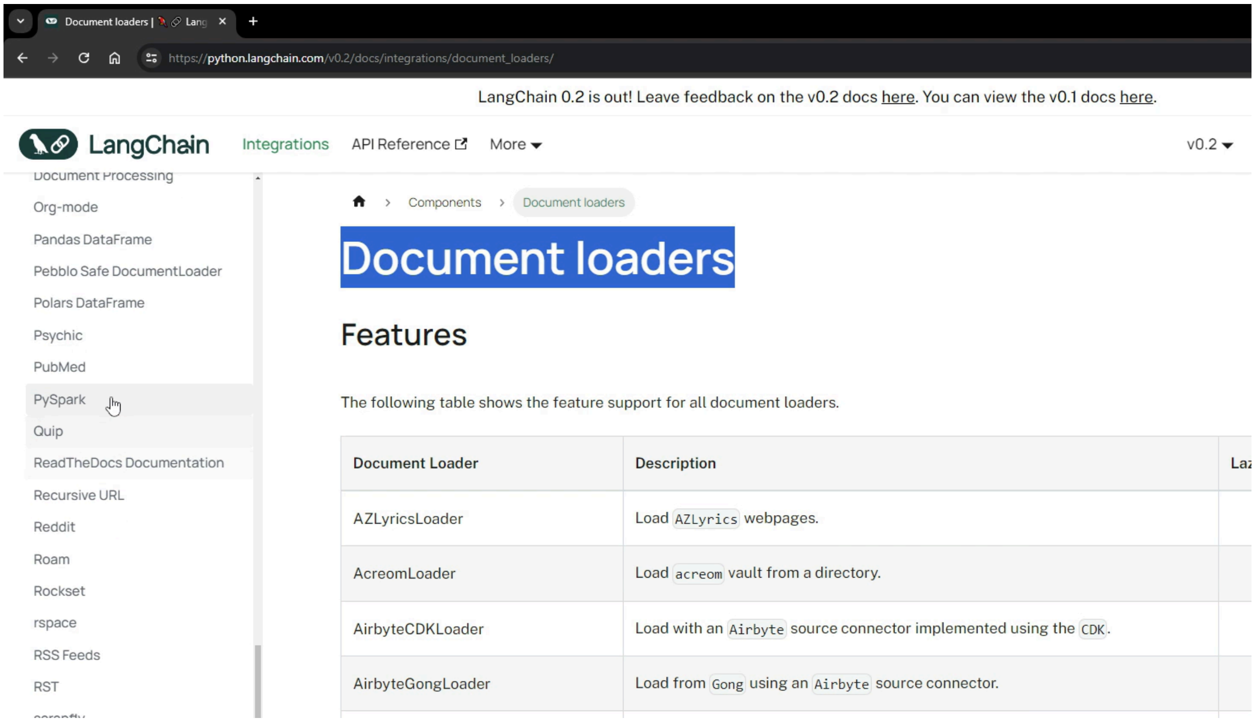
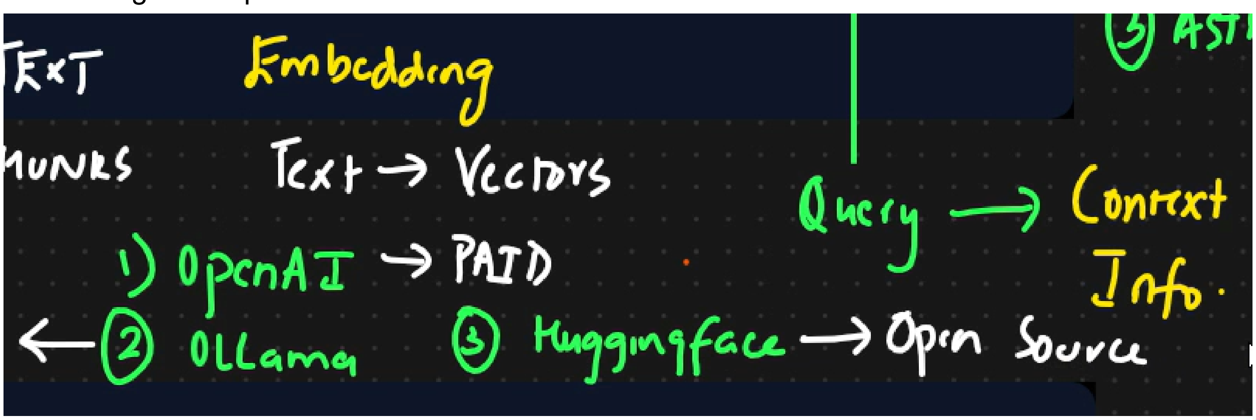

Important Components Of Langchain
① RAG ÷ Retrieval Augmented Generation
DATA SOURCES
LOAD
DATA TRANSFORMATION
SPLIT
EMBED
VectorStore DB
STORE
JSON
URLs
[0.3, 0.4, 0.1, 1.8, 1.1...]
[0.7, 1.4, 2.1, 4.8, 4.1...]
[1.2, 0.3, 1.2, 4.1, 1.8...]
Data Ingestion
Langchain
DATA → TEXT
CHUNKS
Embedding
Text → Vectors
Query → Context
Info.
DATA SOURCE → PDFs
Similarity
Search
Gen AI
Query
LLM Models → Limitation
Context
① FAISS
② CHROMADB
③ ASTRADB


RETRIEVE
Question
PROMPT
LLM
Answer
O/P
Respone.
① Retrieval
CHAIN
AI ASSISTANT
+
Context Info
Interface
Vector Store DB

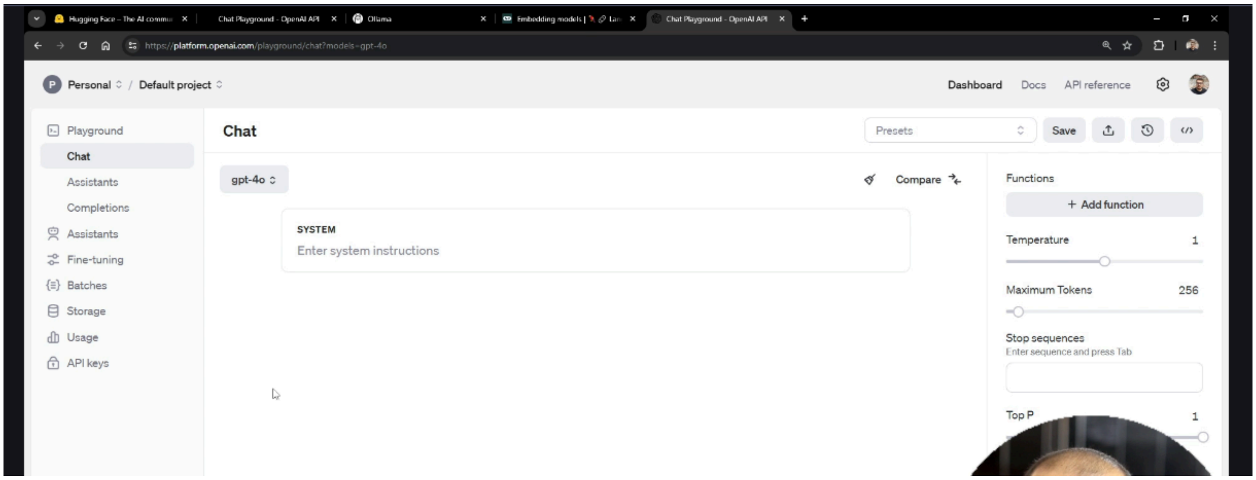
Here are different ways to load documents using langchain. URl changed, but you can research on this.



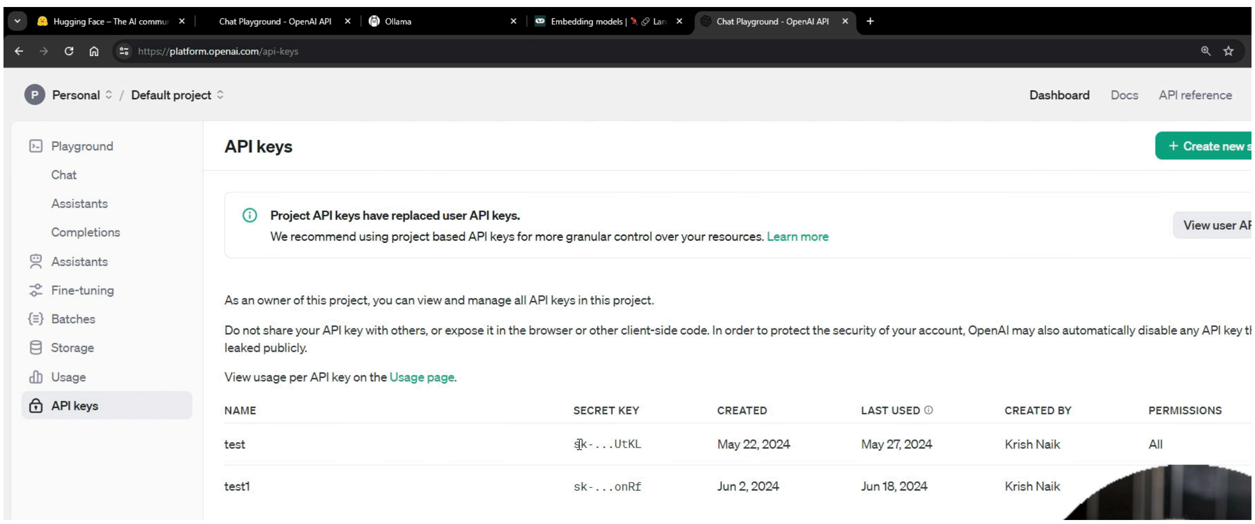
Embedding techniques



For open ai, you have to invest 5 $



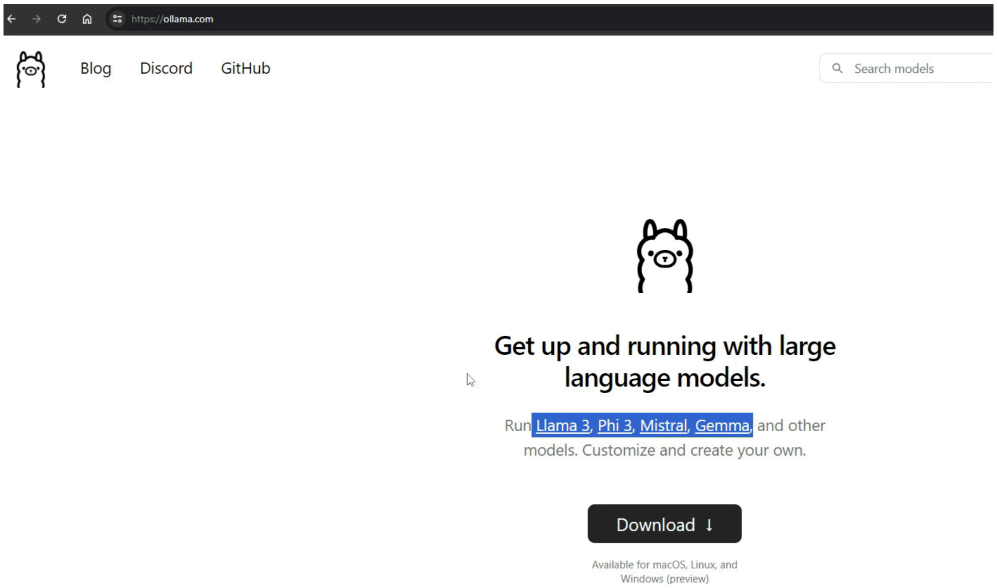
create your own api key



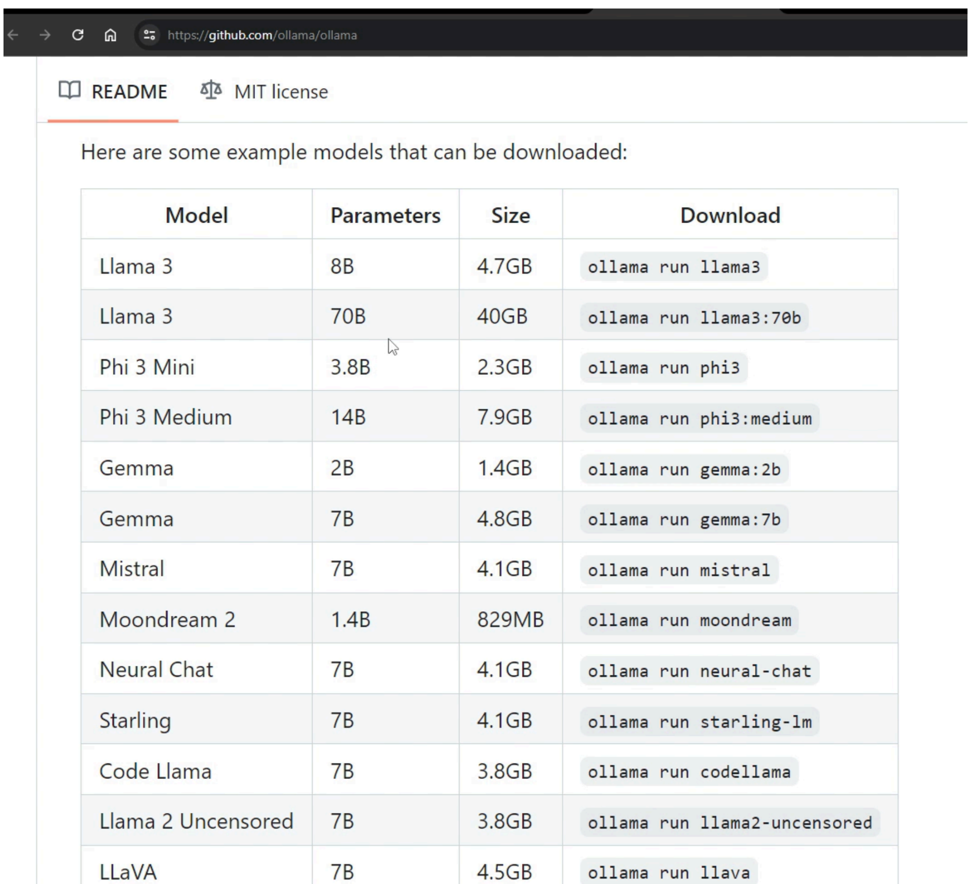
Ollama embedings(refer to the python code)

if you do not want to use openAI api, and you want to use open source models and use embedded techniques is with the platform called ollama

ollama is a platform where you can use open source LLM models.



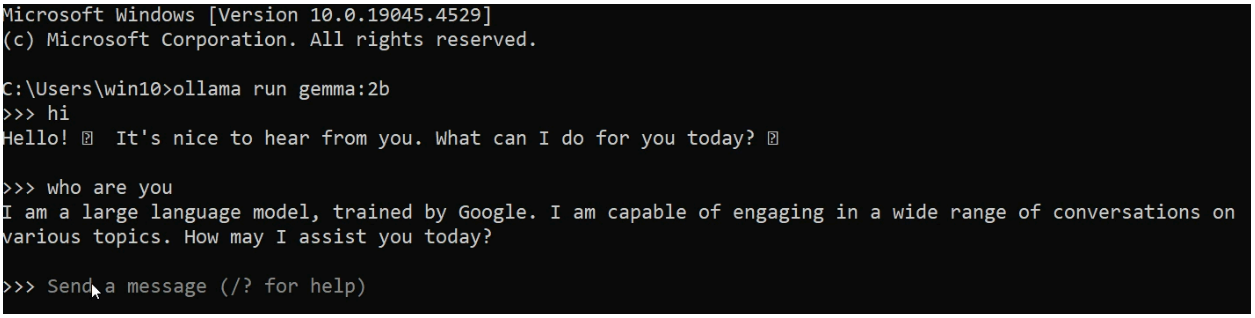
if you go to github of this, it will support all open source models



Here are some example models that can be downloaded:

Model	Parameters	Size	Download
Llama 3	8B	4.7GB	`ollama run llama3`
Llama 3	70B	40GB	`ollama run llama3:70b`
Phi 3 Mini	3.8B	2.3GB	`ollama run phi3`
Phi 3 Medium	14B	7.9GB	`ollama run phi3:medium`
Gemma	2B	1.4GB	`ollama run gemma:2b`
Gemma	7B	4.8GB	`ollama run gemma:7b`
Mistral	7B	4.1GB	`ollama run mistral`
Moondream 2	1.4B	829MB	`ollama run moondream`
Neural Chat	7B	4.1GB	`ollama run neural-chat`
Starling	7B	4.1GB	`ollama run starling-lm`
Code Llama	7B	3.8GB	`ollama run codellama`
Llama 2 Uncensored	7B	3.8GB	`ollama run llama2-uncensored`
LLaVA	7B	4.5GB	`ollama run llava`

after you download it, you can select the model from the list and run it. for example



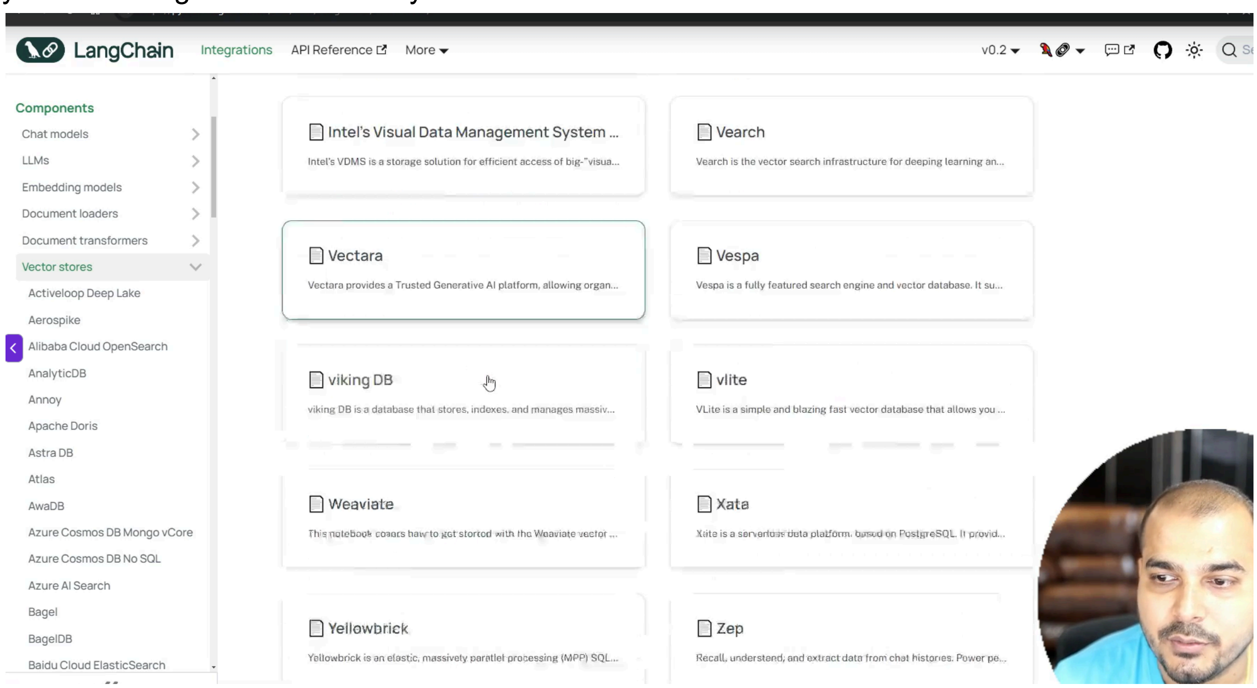
you download another model also for example

Hugging face embeddings

go to huggingface.co and create an account

Here create your access token for accessing the DB

you can also get a vectorDB of your choice from here



Here is the link for other DB's
Chroma
Chroma is a AI-native open-source vector database focused on developer productivity and happiness. Chroma is licensed under Apache 2.0.

https://python.langchain.com/v0.2/docs/integrations/vectorstores/